Claude Code Master Prompt: Document-to-Schema (D2S) Ingestion Engine

Project Background & Context

- **Domain:** Petrochemical & Heavy Industry Enterprise Asset Management.
- **History:** The user previously designed instrument systems and developed an SAP-integrated web platform for asset management. In 2021, 'data cleansing' was viewed as low-ROI maintenance.
- **Current State (2026):** Data cleansing is now a critical enabler (infrastructure) for AI-driven asset management and Digital Twin operations. Accurate specifications must be ingested into the Master DB.

Definition of 'Data Cleansing' for this Project

- Redefining the asset classification hierarchy (Category, Class, Type) based on ISO 14224, tailored to field requirements.
- Defining the 'Master Data' attributes at the lowest hierarchy level (specifications, installation date, asset grade, registration date, asset value, location).
- Applying an algorithm to determine 'Asset Grade' based on maintenance history, qualitative importance, and asset value.
- Populating this defined schema using unstructured sources: legacy datasheets, manual field logs, and inspection reports.

Core Pain Points & Problem Statement

- **Manual Bottleneck:** Even after initial cleansing, ongoing updates require manual human data entry.
- **Data Degradation (Garbage In):** Differing opinions on data importance and lazy input practices lead to polluted, incomplete data over time.
- **Loss of Trust:** The dataset loses reliability, causing field workers to revert to manually checking original paper/PDF datasheets.
- **Technological Shift:** As new technologies emerge (e.g., smart positioners in control valves), legacy data structures become obsolete.

The AI Solution (Enablers & Implementation Ideas)

- **Vision AI & VLM:** Resolves the historical hurdle of scanning and parsing hardcopies/PDFs.
- **Generative AI:** Handles auto-normalization (e.g., unit conversions) and optimizes the display of core information across classification hierarchies.

- **UI/UX Flow:**
  1. User drag-and-drops a datasheet (PDF/Image/Excel).
  2. System auto-identifies the target asset via its Tag number.
  3. AI extracts and normalizes core information, mapping it to the target DB schema.
- **System Integration:** Seamlessly integrates with the existing Drawing Management System and the Asset Integration Platform.
- **Human-in-the-Loop (HITL):** AI updates the staging area; a human MUST click 'Approve' to finalize the DB update.

Execution Directives & Constraints for Claude Code

- **Primary Goal:** Iteratively fill out and complete the 'HD AI/AX Business Problem Definition Workbook'.
- **Required Technologies:** The solution architecture must explicitly incorporate Computer Vision/VLM and Generative AI. Reinforcement Learning is optional but recommended if applicable (e.g., for optimizing the Asset Grade algorithm).
- **Continuous Improvement Architecture:** Design a folder/project structure that supports continuous execution, validation, feedback, and improvement (Harness/Loop structure).
- **Logging & Memory:**
  - Maintain a strict change log of all modifications made by the user and AI.
  - Store critical insights and strategic decisions in a dedicated memory file.

Folder Structure & Initialization Plan

```
/project_root
  /docs
    - HD_AIAX_Workbook.md (The target document to be filled)
    - change_log.md (Tracks all user/AI modifications)
    - insight_memory.md (Stores key decisions and ideas)
  /src
    - ingestion_pipeline.py (Vision/LLM processing)
    - schema_mapping.py (Data standardization)
  /tests
    - test_pipeline.py
```

Action Request for Claude Code

Acknowledge these instructions. Initialize the required folder structure and logging files. Then, await further input from the user to begin filling out the HD AI/AX Workbook step-by-step.